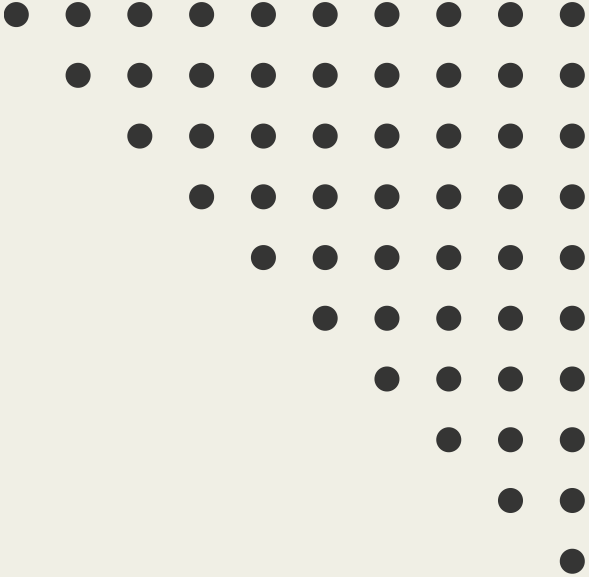
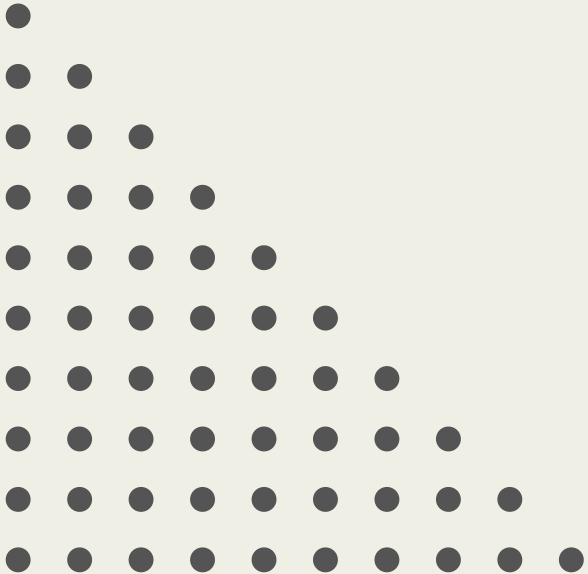




MINI PROJECT

LEGALLYBOUND

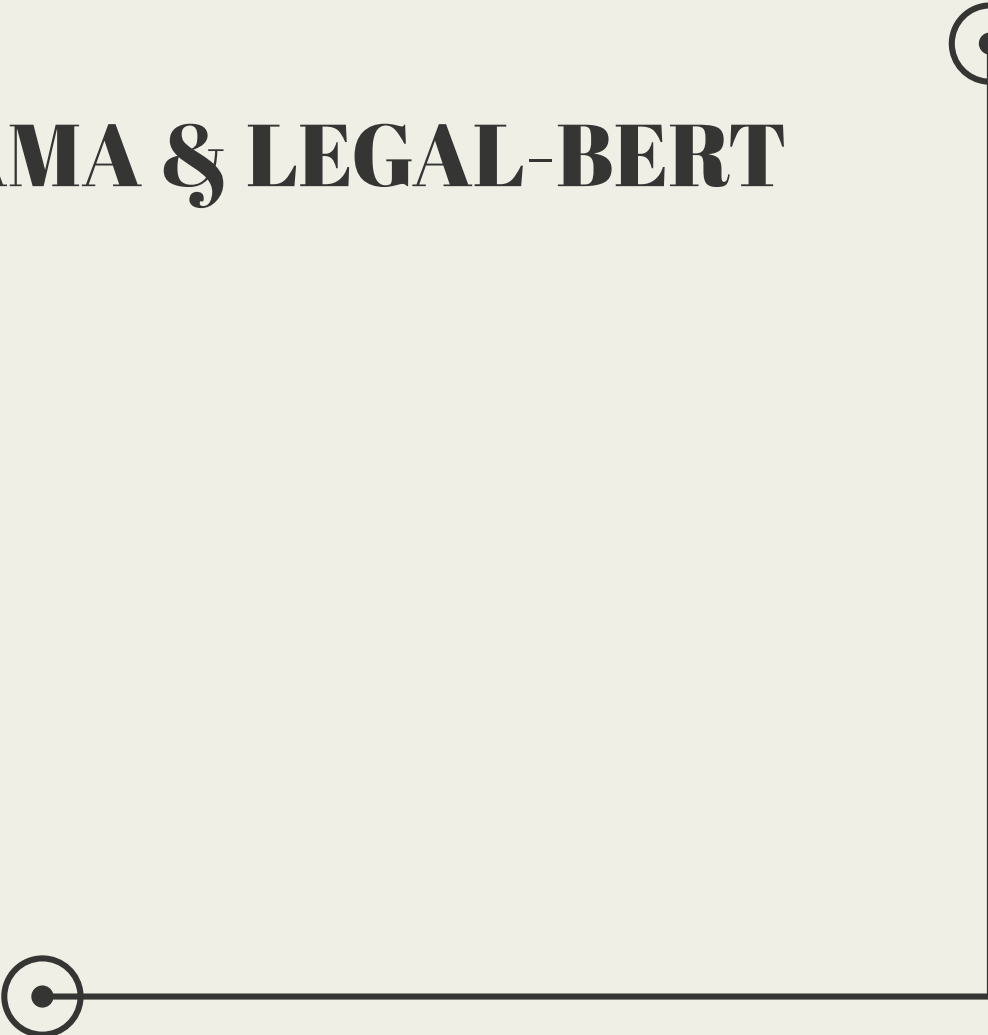
AI-POWERED CONTRACT RISK ANALYSIS USING TINYLLAMA & LEGAL-BERT

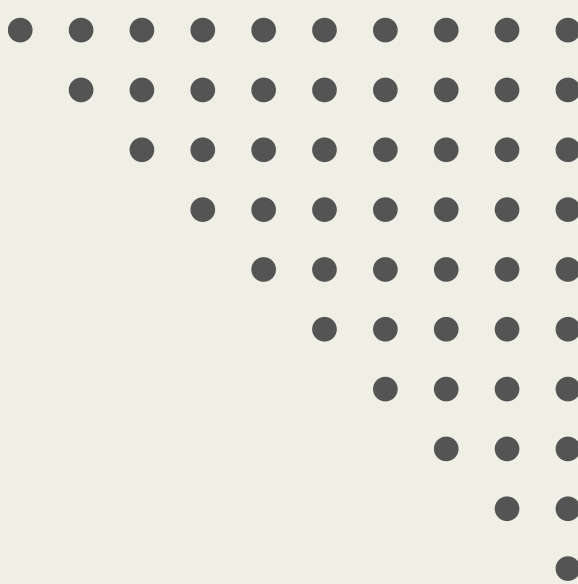


By

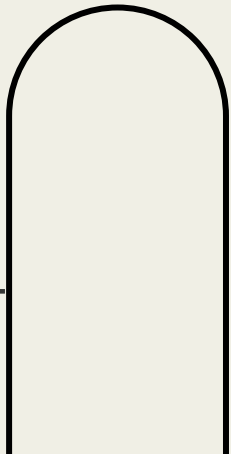
Amod Adarsh – CS23B1005

Deva anand M– CS23B1020





PROBLEM STATEMENT

- **LEGAL CONTRACT REVIEW IS MANUAL, SLOW, AND EXPENSIVE — LAWYERS CHARGE \$200–\$500/HR**
 - **A SINGLE COMMERCIAL CONTRACT HAS 30–50 CLAUSES THAT NEED RISK ASSESSMENT**
 - **SMALL BUSINESSES & STARTUPS OFTEN SIGN CONTRACTS WITHOUT FULL LEGAL REVIEW**
 - **MISSING A HIGH-RISK CLAUSE CAN LEAD TO MILLIONS IN EXPOSURE**
- 
- 
- 
- 

TECH STACK

Component	Technology	Purpose
Clause Extraction	TinyLlama 1.1B	Identify & summarize clauses
Risk Classification	Legal-BERT (fine-tuned, 3-class)	HIGH / MEDIUM / LOW labeling
Embeddings	all-MiniLM-L6-v2 (384-dim)	Semantic similarity
Vector Store	DuckDB + VSS (HNSW)	Storage & retrieval
Risk Labels	Groq API (Llama 3.1 8B)	Training data generation
Dataset	CUAD v1 (500+ contracts, 41 clause types)	Training corpus
Backend	FastAPI + Uvicorn	REST API + SSE streaming
Frontend	Vanilla HTML/CSS/JS	Dark-mode premium UI

INPUT

CUAD.JSON

"text": "<|system|>\nYou are an expert legal contract assistant. Answer legal contract questions with meaningful legal text and keywords.

</s>\n<|user|>\nIs there a requirement for insurance that must be maintained by one party for the benefit of the counterparty?

</s>\n<|assistant|>\nIn partial spend of the fixed benefits allowance referred to in sub-clause 6.1, the Executive and his spouse or civil partner and any children under the age of 21 (or 24 if in full time education) are entitled to membership of a private medical insurance scheme.</s>"

INPUT DATA SPLIT

```
cuad.json contracts : 510
Downloading master_clauses.csv...
CSV shape           : (510, 83)
From CSV             : 4505
Total combined      : 9833
40% sample          : 3933
Train                : 2753
Validation           : 589
Test                 : 591
```

VOCAB SIZE: 32001

```
=====
INPUT DIMENSION FULL ANALYSIS
=====

Single Sample Shapes:
  input_ids      : [256]  -> torch.int64
  attention_mask : [256]  -> torch.int64
  labels         : [256]  -> torch.int64

Batched Shape (batch_size = 1):
  input_ids      : [1, 256]
  attention_mask : [1, 256]
  labels         : [1, 256]

Token Breakdown:
  MAX_LENGTH      : 256
  Real tokens     : 130
  Pad tokens      : 126
  Prompt tokens (-100) : 182
  Supervised tokens : 74

Model Output Shapes:
  logits          : [1, 256, 32001]
  shift_logits    : [1, 255, 32001]
  shift_labels    : [1, 255]
  ...
```


LOSS FUNCTION AND ACTIVATION

Causal Language Modeling Loss

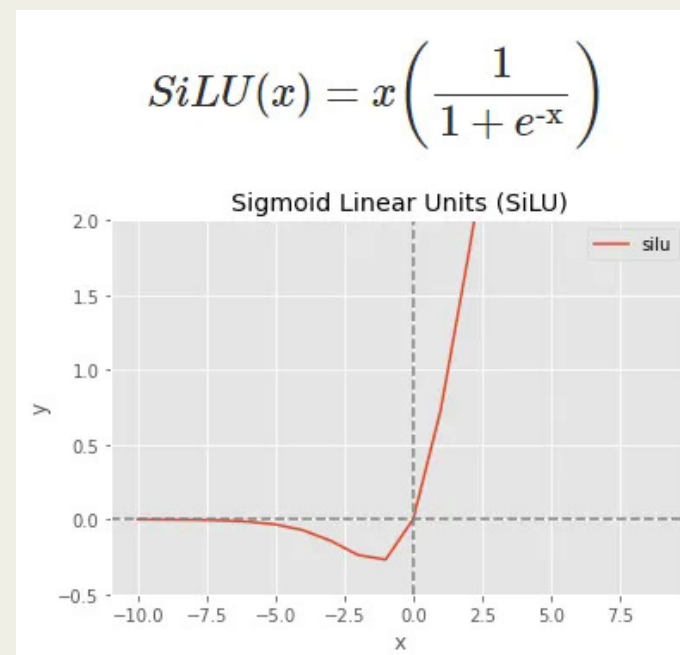
$$L_{CLM}^{(x)} = -\frac{1}{|x|} \sum_{i=1}^{|x|} \log P(x_i / x_{<i})$$

where:

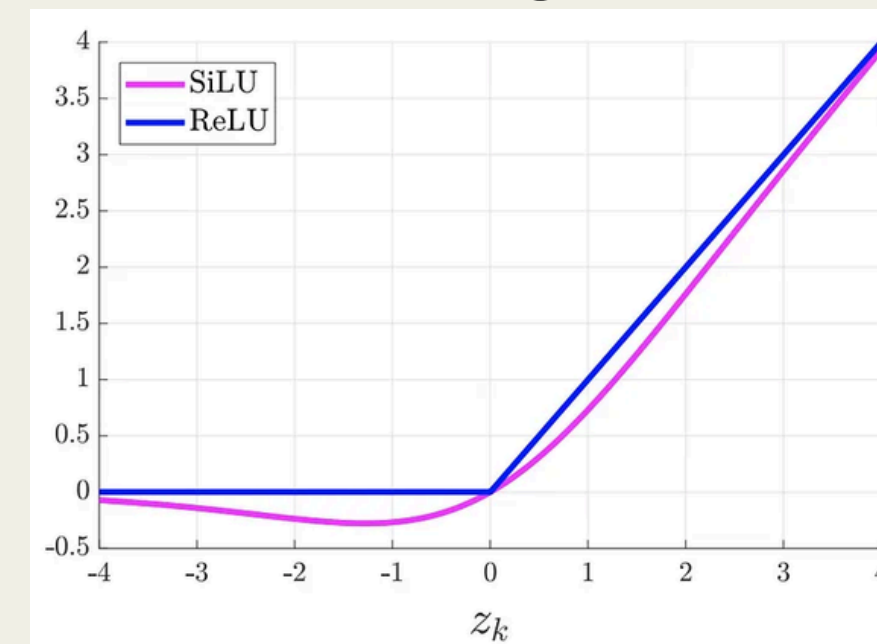
$x = \{x_1, x_2, x_3, \dots, x_{|x|}\}$ represents a sequence

$x_{<i} = x_1, x_2, x_3, \dots, x_{i-1}$

Activation Function :



Addresses “Dying ReLU”



EXPERIMENTAL DESIGN

EXPERIMENT	LAYERS TRAINED
Baseline	None
Exp 1	Spread (1,7,14,20,22)
Exp 2	Last 10 (13–22)
Exp 3	Last 5
Exp 4	Layer 1
Exp 5	Layers 1 & 2
Exp 6	Alternating Odd
Exp 7	Bottom 3 + Top 2

BASELINE

```
Both `max_new_tokens` (=100) and `max_length` (=
...

```

Notes:

- *UNEXPECTED* :can be ignored when Loading fr
- *MISSING* :those params were newly initia

Output is truncated. View as a [scrollable element](#) or open in

Zero-shot samples : 30

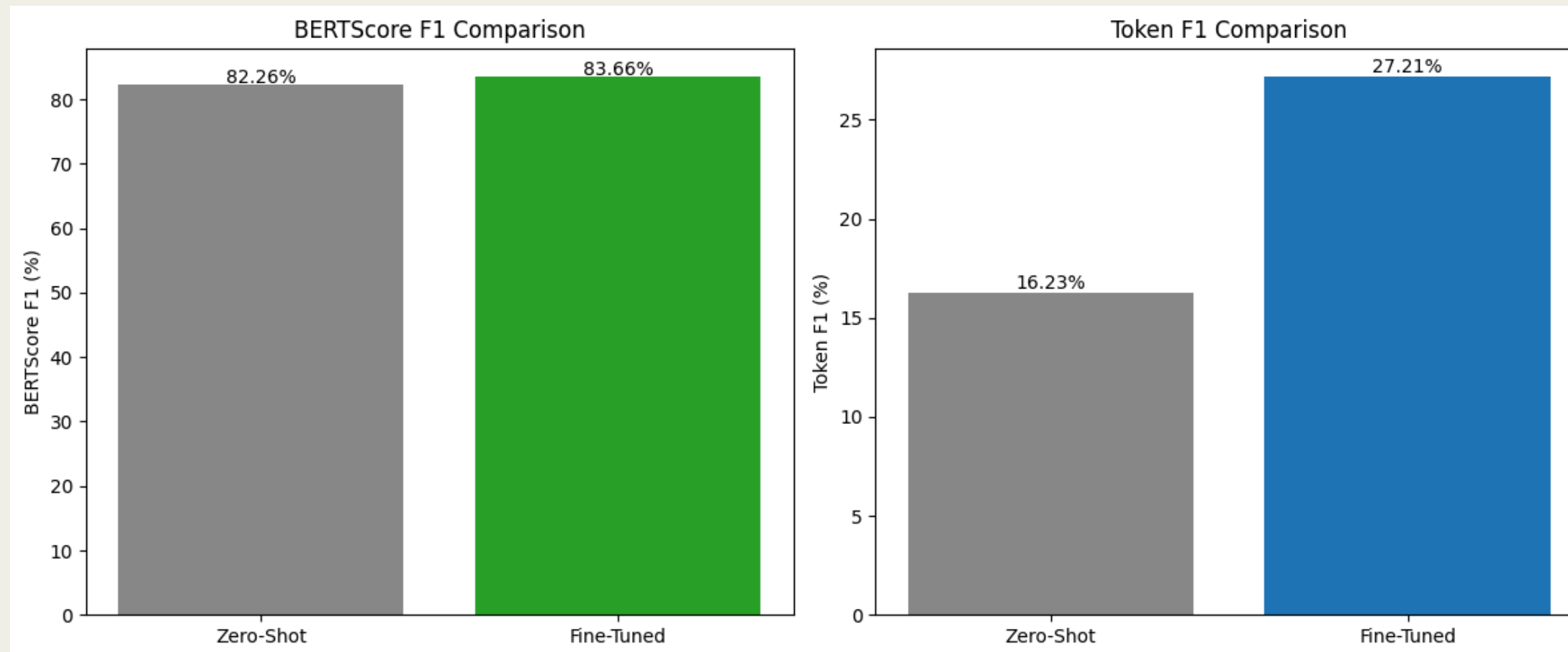
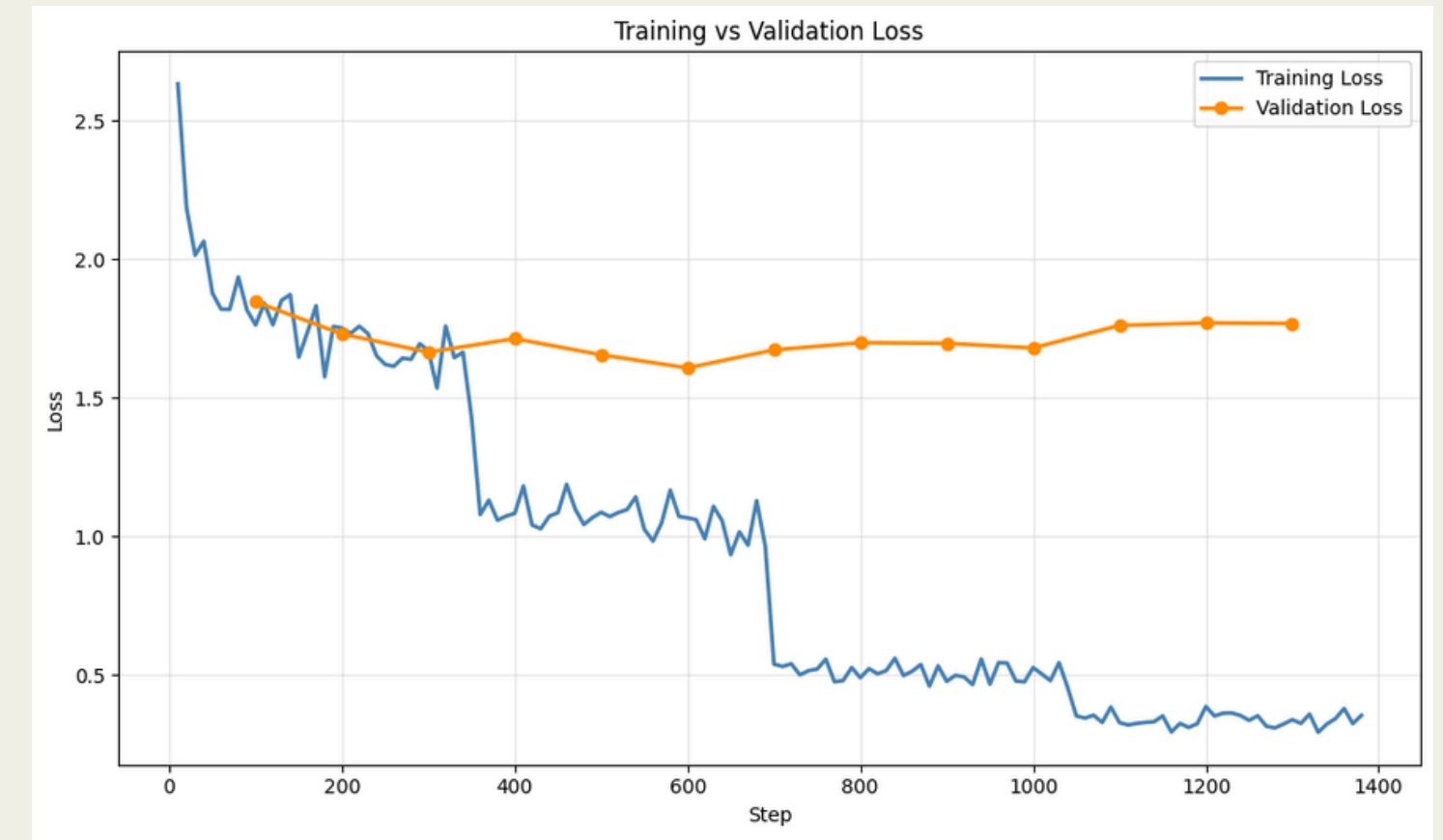
Zero-shot BERTScore F1: 82.26%

Zero-shot Token F1 : 16.23%

Zero-shot baseline done. GPU memory cleared.

SPREAD (1,7,14,20,22)

- Layers Trained: 1, 7, 14, 20, 22
- Rationale:
 - Layer 1: Input processing (first contact with legal text)
 - Layer 7: Mid-level features
 - Layer 14: Deep semantic understanding
 - Layers 20, 22: Output generation
- Trainable params : 25 % of total parameter approx
- steps: 1380 : Epoch 4
- First experiment with skip-pattern training. This is the concept of distributed fine-tuning.



LAST 10 (13–22)

- Layers Trained: 13, 14, 15, 16, 17, 18, 19, 20, 21, 22

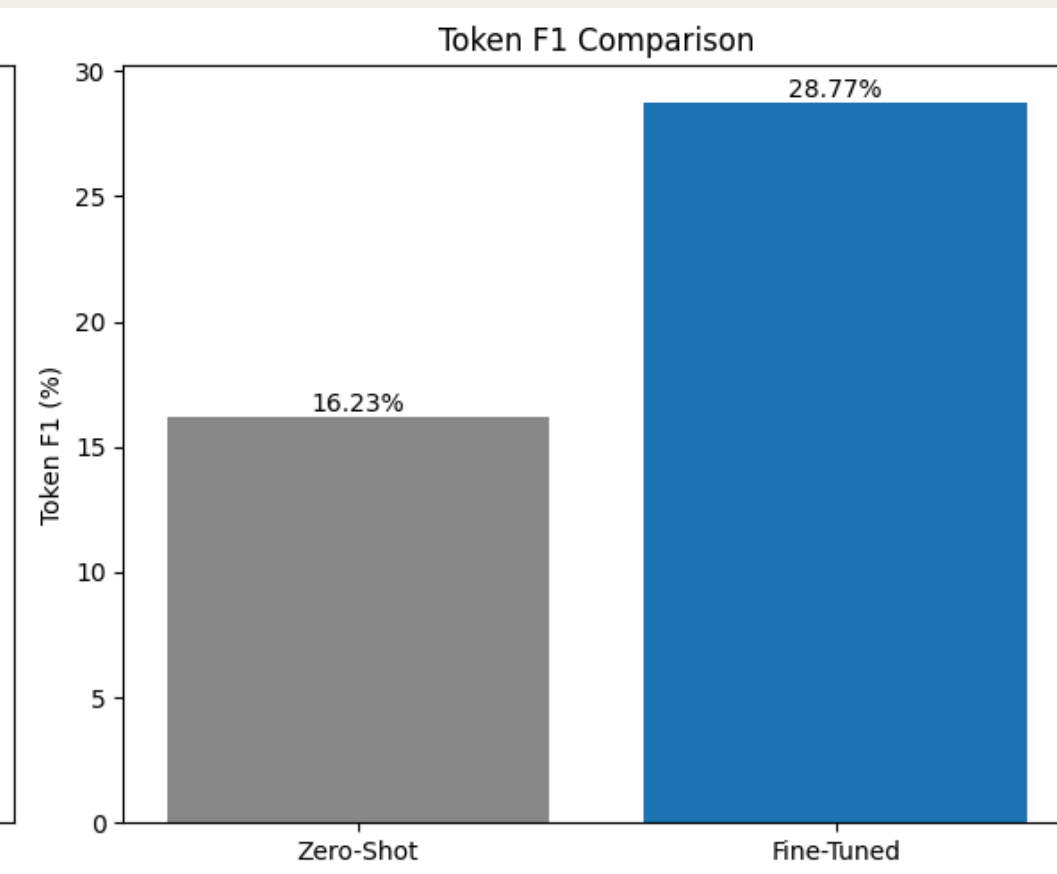
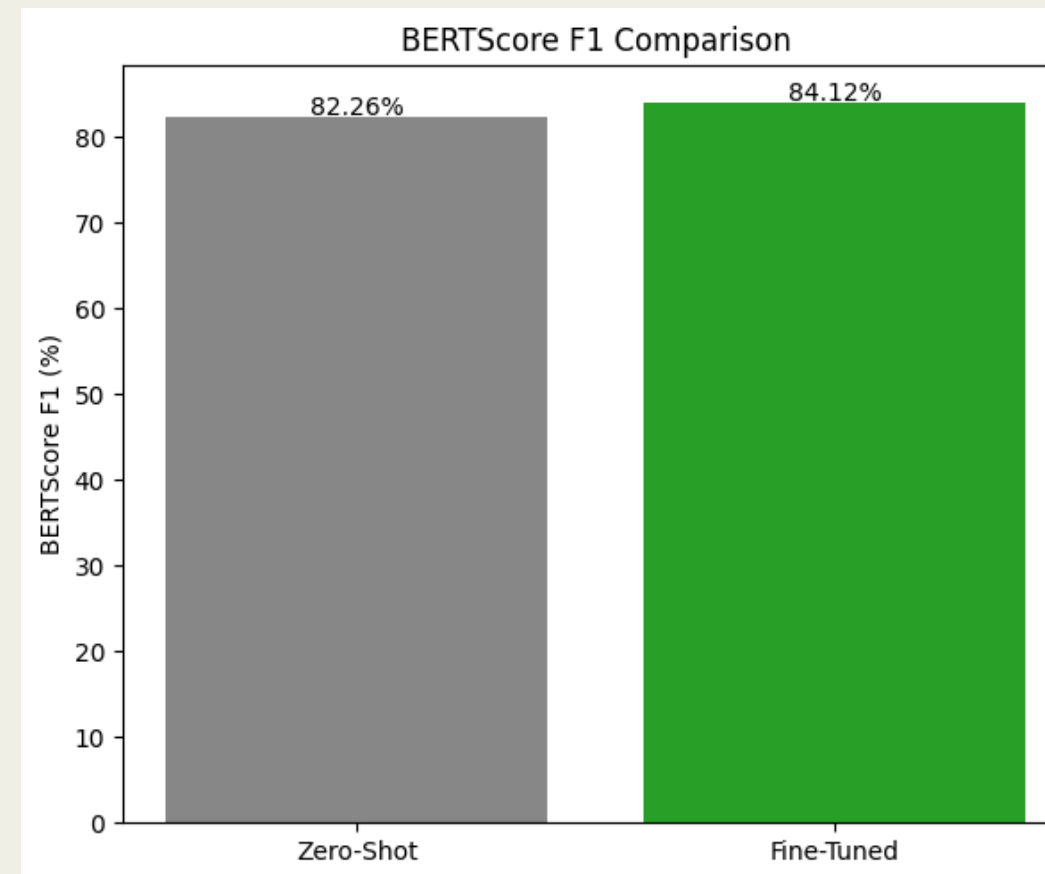
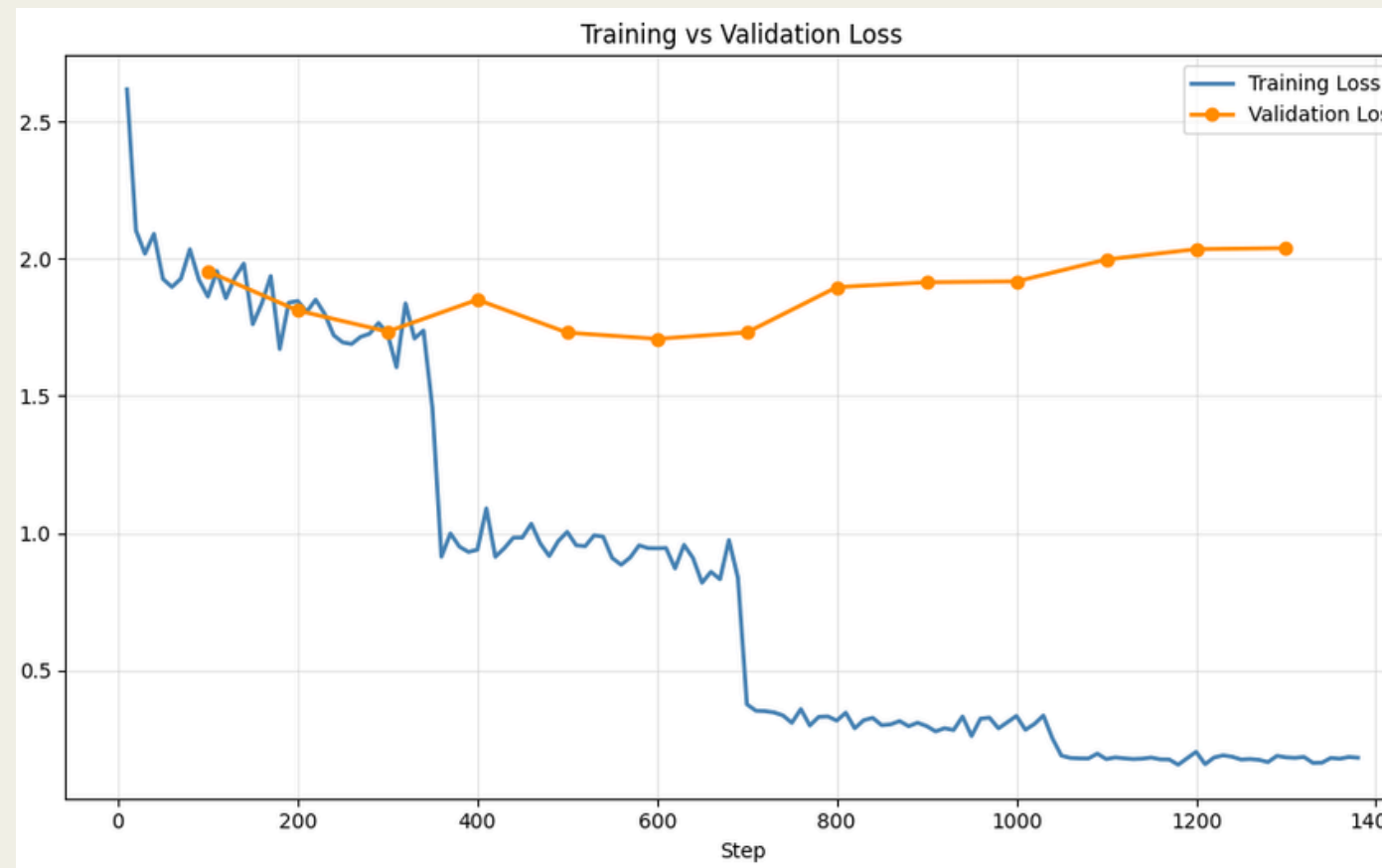
- Rationale:

The second half of the network handles task-specific reasoning. Training all 10 layers should maximize legal understanding.

- Trainable params : 45 % of total parameter approx
- steps: 1380 : Epoch 4
- Training more layers improved semantic understanding but caused severe overfitting.

Summary:

```
{
  "zero_shot_bert_f1": 82.26,
  "zero_shot_token_f1": 16.23,
  "fine_tuned_bert_f1": 84.12,
  "fine_tuned_token_f1": 28.77,
  "train_token_accuracy": 85.64,
  "validation_token_accuracy": 64.2,
  "test_token_accuracy": 65.19,
  "final_train_loss": 0.18154237270355225,
  "final_validation_loss": 2.0389161109924316,
  "final_test_loss": 1.6403,
  "training_loss_below_validation_loss": true,
  "beats_zero_shot": true,
  "model_dir": "tinylama_layers_13_22_masked",
  "latest_checkpoint": "output_layers_13_22_masked\\checkpoint-1380"
}
```



COMBINATION

LAYER 1

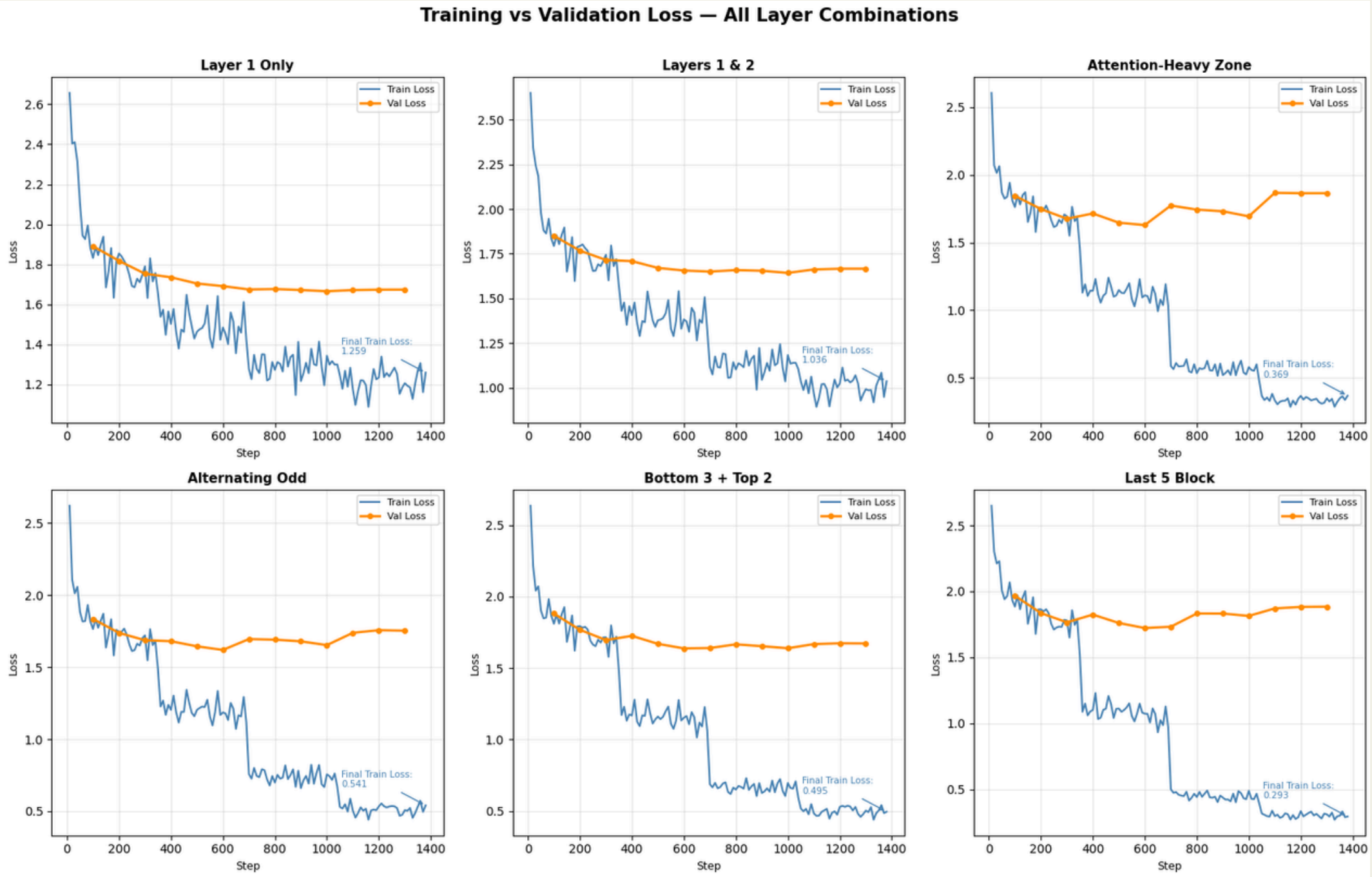
- Trained only Layer 1 — the very first transformer block
- Rationale:
 - Layer 1: Input processing (first contact with legal text)
- Trainable params : 4% of the entire model
- steps: 1380 : Epoch 4
- Layer 1 is where raw word tokens get their first contextual transformation. By fine-tuning just this layer, we are teaching the entire model to perceive legal text differently from the very first step. All downstream layers inherit this legal awareness

LAST 5 LAYERS

- Trained : 18, 19, 20, 21, 22
- Trainable params : 26% of the entire model
- steps: 1380 : Epoch 4
- Each layer keeps shifting the feature distribution for the next, so the optimizer can't find a consistent direction, resulting in poorer convergence.

LAYER 1&2

- Training the first two layers together should create a stronger legal input processor.
- Trainable params : 8% of the entire model
- steps: 1380 : Epoch 4
- Adding layer 2 actually REDUCED BERTScore compared to layer 1 alone (84.19% → 83.84%). Due to **gradient interference** between neighboring layers.



COMBINATION

ALTERNATING ODD [1,3,5,7,9]

- Trained - 1,3,5,7,9
- Rationale:

Layer 1 → Legal vocabulary recognition
(frozen layer 2 preserves this)

Layer 3 → Phrase-level legal patterns
(frozen layer 4 preserves this)

Layer 5 → Clause-level understanding
(frozen layer 6 preserves this)

Layer 7 → Contract structure
(frozen layer 8 preserves this)

Layer 9 → Deep legal reasoning

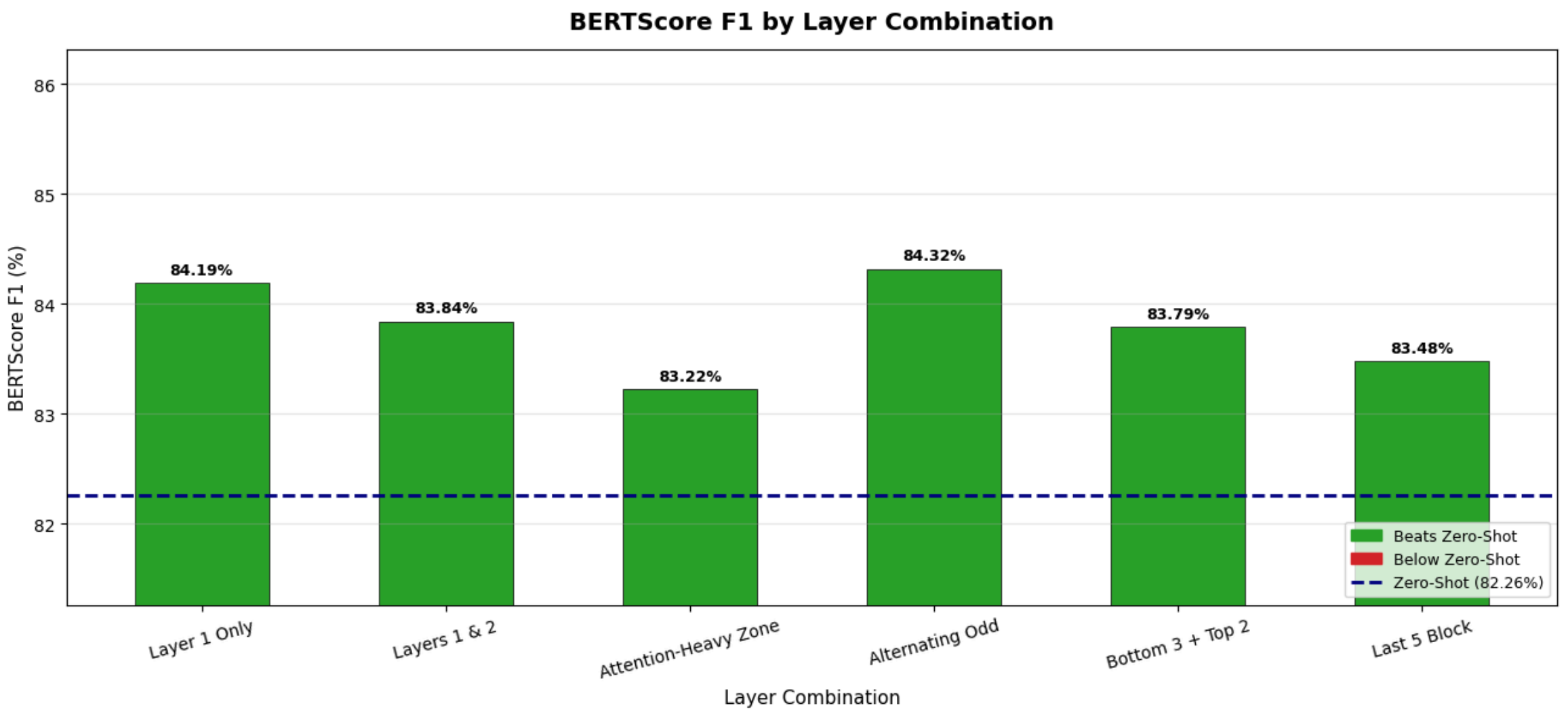
- Trainable params : 26% of the entire model
- steps: 1380 : Epoch 4
- Each trained layer specializes on one aspect of legal understanding, and frozen layers preserve those specializations as information flows forward. No gradient interference, maximum specialization.

BOTTOM 3 + TOP 2

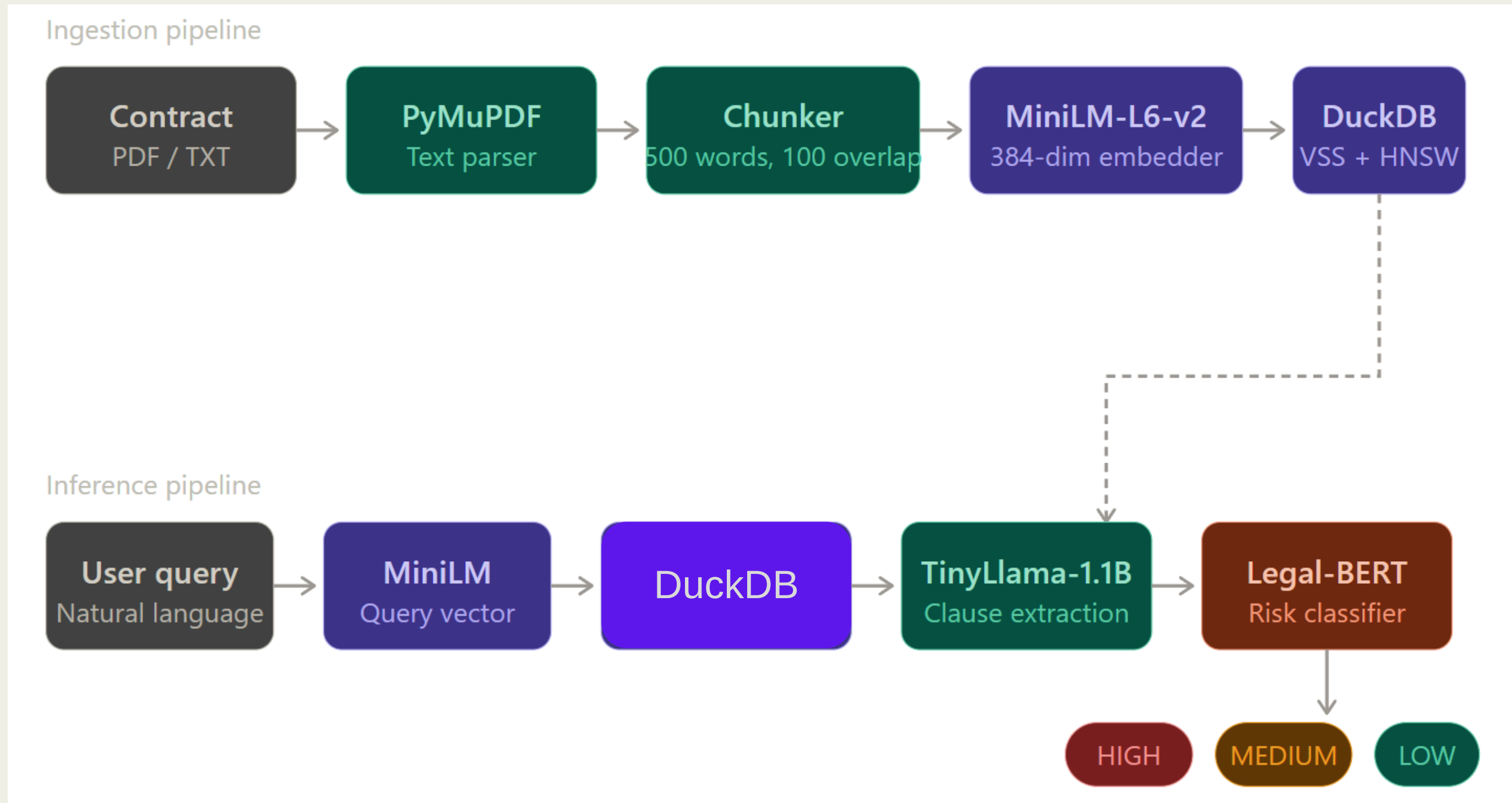
- Trained -1,2,3,21,22
- Rationale:

Train the early layers to adapt legal vocabulary AND train the final layers to improve legal text generation, while keeping the entire reasoning middle frozen.

- Trainable params : 22% of the entire model
- steps: 1380 : Epoch 4
- The early layers learn new representations, the late layers expect a different feature distribution, and the frozen middle layers cannot adapt to align these mismatched representations.



SYSTEM ARCHITECTURE



OUTPUT SCREENSHOTS - RISK EXTRACTION

● General 🗑 36% LOW • 92% Exhibit 4.5 CONSULTING AGREEMENT - DR. GAETANO MORELLO N.D. INC. This Agreement is made effective on the 10th day of January 2019 (the "Effective Date"). BETWEEN: EMERALD HEALTH NATURALS, INC. Registered and Records Office: 7860 Venture Street, Burnaby, BC V5A 1V3 (the "Company") AND: DR. GAETANO MORELLO N.D. INC. Bus 💡 The given clause is at a risk level of LOW, which means it is not a critical risk that should be addressed immediately. The clause is related to the Exhibit 4.5 CONSULTING AGREEMENT, which is a contract between the Company and Dr. Gaetano Moreello N.D. Inc. (the "Consultant"), and outlines the terms and conditions of the consulting agreement. The Consultant is providing consulting services to the (confidence: 92%) ▶ Show excerpt Exhibit 4.5 CONSULTING AGREEMENT - DR. GAETANO MORELLO N.D. INC. THIS AGREEMENT made effective the 10 day of January 2019 (the "Effective	● Payment Terms 🗑 40% MEDIUM • 100% The Company agrees not to deduct or retain any amounts from the Fees due to the Contractor, nor shall it be obliged to remit same to the required governmental authority, any amounts that may be required by law or regulation to be deducted, retained and remitted including, without limitation, Federal and Provincial or State Income Tax, Workers' Compensation and Pension Plan deductions and remittances. All such remittances and other payments are entirely the Company's responsibility. 💡 The given clause is classified at the MEDIUM risk level because it requires the Company to pay the Contractor in full, without deducting or retaining any amounts, and to not be required to remit any amounts to the governmental authority or required by law or regulation to be deducted, retained, and remitted. This risk level is appropriate for a contract where the Company is responsible for paying the Contractor, and the Contractor is not required to deduct (confidence: 100%) ▶ Show excerpt the Company 3.3 Deductions and Remittances. Emerald shall not be obliged to deduct or retain from the Fees due to the Contractor, nor shall
---	--

● Subcontractor Agreement

📄 38%

MEDIUM • 100%

The contractor agrees to provide services to the client as a subcontractor, and the client agrees to pay the contractor for those services. The contractor represents that it has sought independent legal advice or has chosen not to do so with full knowledge of the risks associated with not obtaining such independent legal advice. The contract also states that time is of the essence and that any assignment without the consent of the other party is prohibited.

💡 The given clause is classified at a risk level of MEDIUM due to the following reasons: 1. The clause relates to a subcontractor agreement, which is a contractual relationship between a contractor and a client. The subcontractor's services are essential to the client's project, and the client is relying on the contractor to provide these services. The contractor's representation that it has sought independent legal advice or has chosen not to do (confidence: 100%)

► Show excerpt

of ensuring its, his or her rights and interests are protected. Each party to the Agreement represents to the other that it, he or she has

● Cause

📄 40%

MEDIUM • 100%

"Cause" includes gross negligence or gross incompetence in the conduct of the Contractor's Services, material breach of any term of the Agreement, and failure to remedy such breach within 60 days after written notice.

💡 The given clause is classified at the MEDIUM risk level due to the potential for gross negligence or gross incompetence in the conduct of the Contractor's Services, material breach of any term of the Agreement, and failure to remedy such breach within 60 days after written notice. This risk level is based on the fact that the Contractor's Services may involve risks of causing harm to the Project or the Client, and the (confidence: 100%)

► Show excerpt

building companies involved in the medical or recreational cannabis industries. (d) "Cause" includes, without limitation, the following: (i)

● Information Related to the Business

🔒 39%

MEDIUM • 100%

This information relates to the Business of the Company, including the Company's ownership of, and its use of, Intellectual Property owned, licensed or developed by or for Emerald.

💡 The given clause is at a risk level of MEDIUM due to the following contextual explanation: The information relates to the Business of the Company, including the Company's ownership of, and its use of, Intellectual Property owned, licensed or developed by or for Emerald. This information is relevant to the Company's business operations and should be considered in the context of the Company's overall business strategy and risk management framework. (confidence: 100%)

▸ Show excerpt

or (ii) his or her relationship with Emerald or an Affiliate of Emerald; which information is not generally known in the industry in which the

● Information Rights

🔒 45%

MEDIUM • 100%

The contractor agrees to provide the client with all information, ideas, concepts, and know-how related to the project, including but not limited to meeting and collaboration notes, contents of laboratory notebooks, data, formulae, drawings, diagrams, blueprints, know-how, concepts, processes, product plans, service plans, computer software, flowcharts, specifications, design documents, and models, as well as business information including data, databases, business models, market research and forecasts, and customer lists. The contractor shall not disclose any such information to any third party without the express written consent of the client.

💡 The given clause is at a risk level of MEDIUM due to the information rights clause. The clause states that the contractor agrees to provide the client with all information, ideas, concepts, and know-how related to the project, including but not limited to meeting and collaboration notes, contents of laboratory notebooks, data, formulae, drawings, diagrams, blueprints, know-how, concepts, processes, product plans, service plans, computer software, (confidence: 100%)

▸ Show excerpt

including meeting and collaboration notes, contents of laboratory notebooks, data, formulae, drawings, diagrams, blueprints, know-how,

OUTPUT VERIFICATION

Clause	Our system	Claude assessment	Verdict
General	LOW, 92%	LOW	Correct
Payment Terms	MEDIUM, 100%	MEDIUM (borderline HIGH)	Mostly correct – the indemnification component is underweighted
Cause	MEDIUM, 100%	MEDIUM	Correct
Information Rights	MEDIUM, 100%	HIGH	Wrong – the IP clause is extremely broad and should be HIGH
Information Related to Business	MEDIUM, 100%	HIGH	Wrong – same issue, this is the confidential information definition which feeds into the IP assignment
Subcontractor Agreement	MEDIUM, 100%	MEDIUM	Correct

SEMANTIC SEARCH

 Upload & Analyze

 Semantic Search

what is the expiry date of this contract

 Search

AI ANSWER

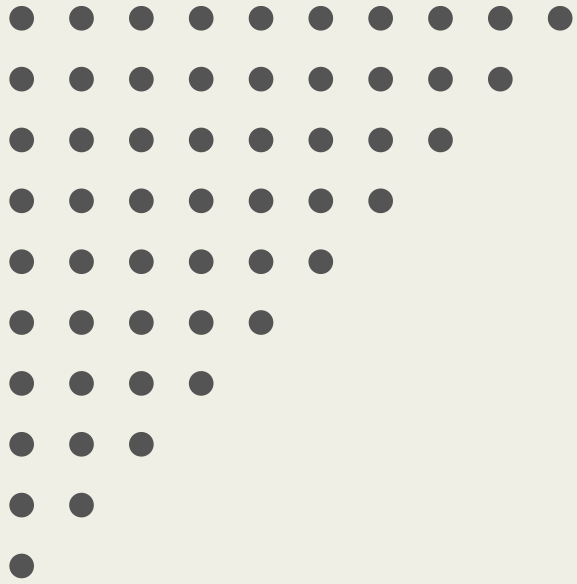
The contract has an expiry date of 30 days after the 10th day of January 2019.

 **LOW** • **97%**

The clause "Expiry date of the contract is 30 days after the 10th day of January 2019" has been classified at this risk level due to the following reasons: 1. The clause is a deadline, which means that it is a critical aspect of the contract. 2. The 10th day of January 2019 is a specific date, which is important for the parties to plan and prepare for the (confidence: 97%)

LIMITATIONS

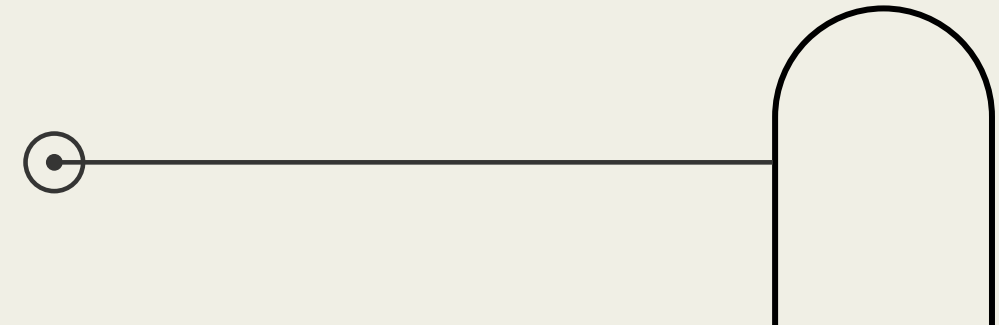
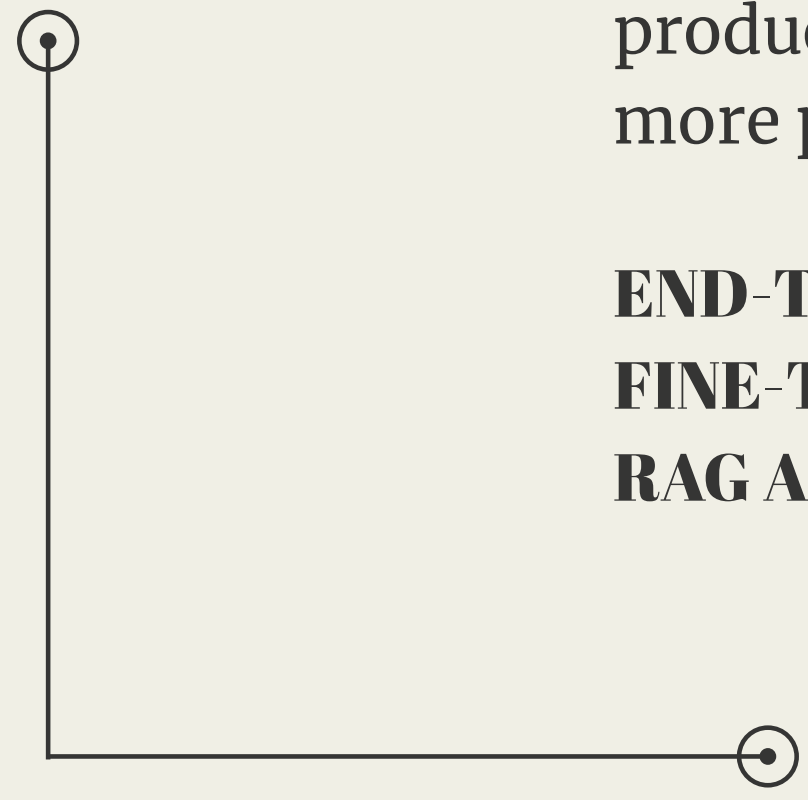
- Risk Extraction - the risk extraction labels are not always accurate
- Context window - 256 tokens max; long clauses get truncated
- Model size - 1.1B params limits reasoning depth vs GPT-4 class models
- Risk labels - generated by LLM (Llama 3.1), not human-verified
- English only - no multi-language support
- Not legal advice - tool assists review, cannot replace lawyers

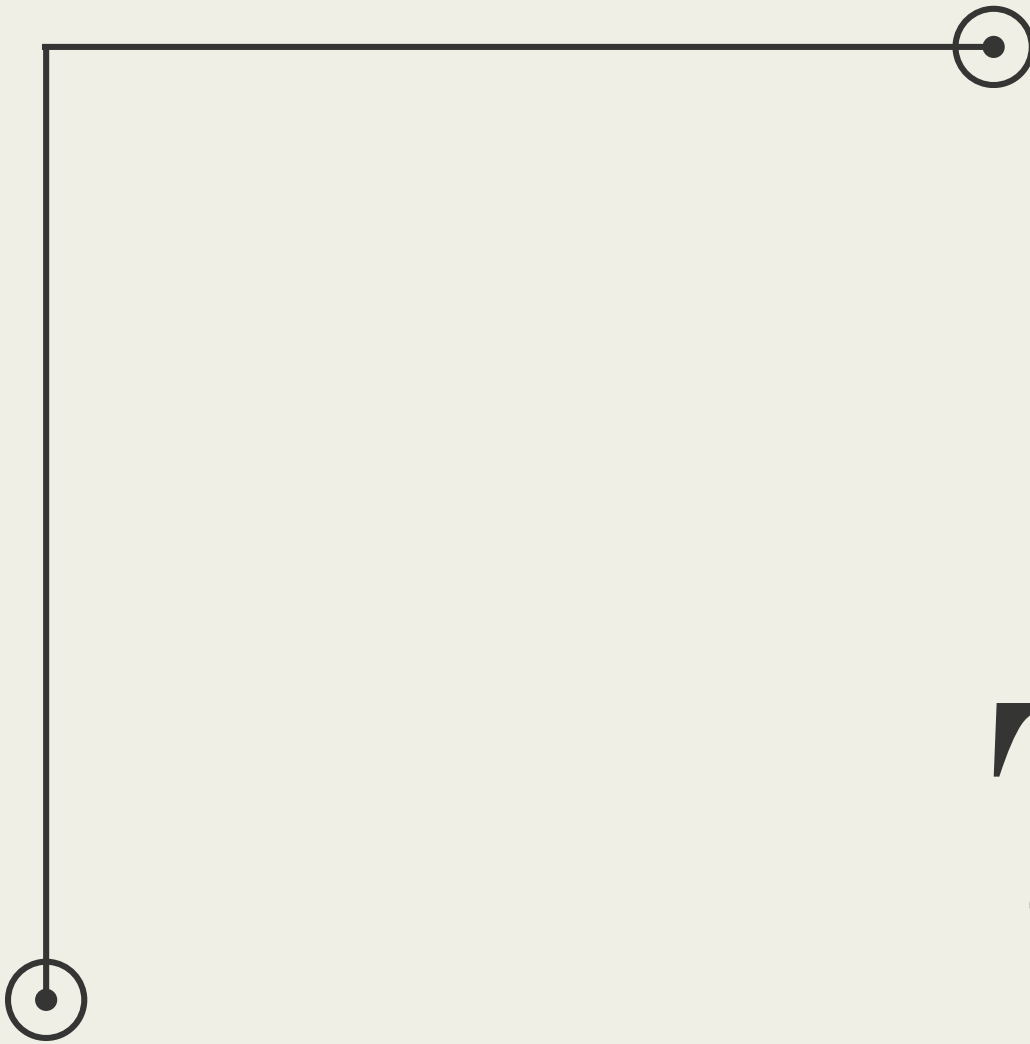


CONCLUSION

The best result came from alternating odd layers [1,3,5,7,9] at 84.32% BERTScore, which worked because frozen layers between trained ones prevented gradient interference and allowed each trained layer to specialize cleanly. More surprisingly, training just Layer 1 alone only 4% of the model achieved 84.19% BERTScore, because adapting the first transformation applied to input tokens creates a legal vocabulary lens that silently improves all 21 downstream frozen layers. Meanwhile, training 10 consecutive layers produced the worst generalization despite the highest training accuracy, confirming that more parameters cause more memorization, not more learning.

END-TO-END CONTRACT RISK ANALYSIS PIPELINE
FINE-TUNED TINYLLAMA + LEGAL-BERT ON CUAD DATASET
RAG ARCHITECTURE WITH DUCKDB VECTOR STORE





THANK YOU

